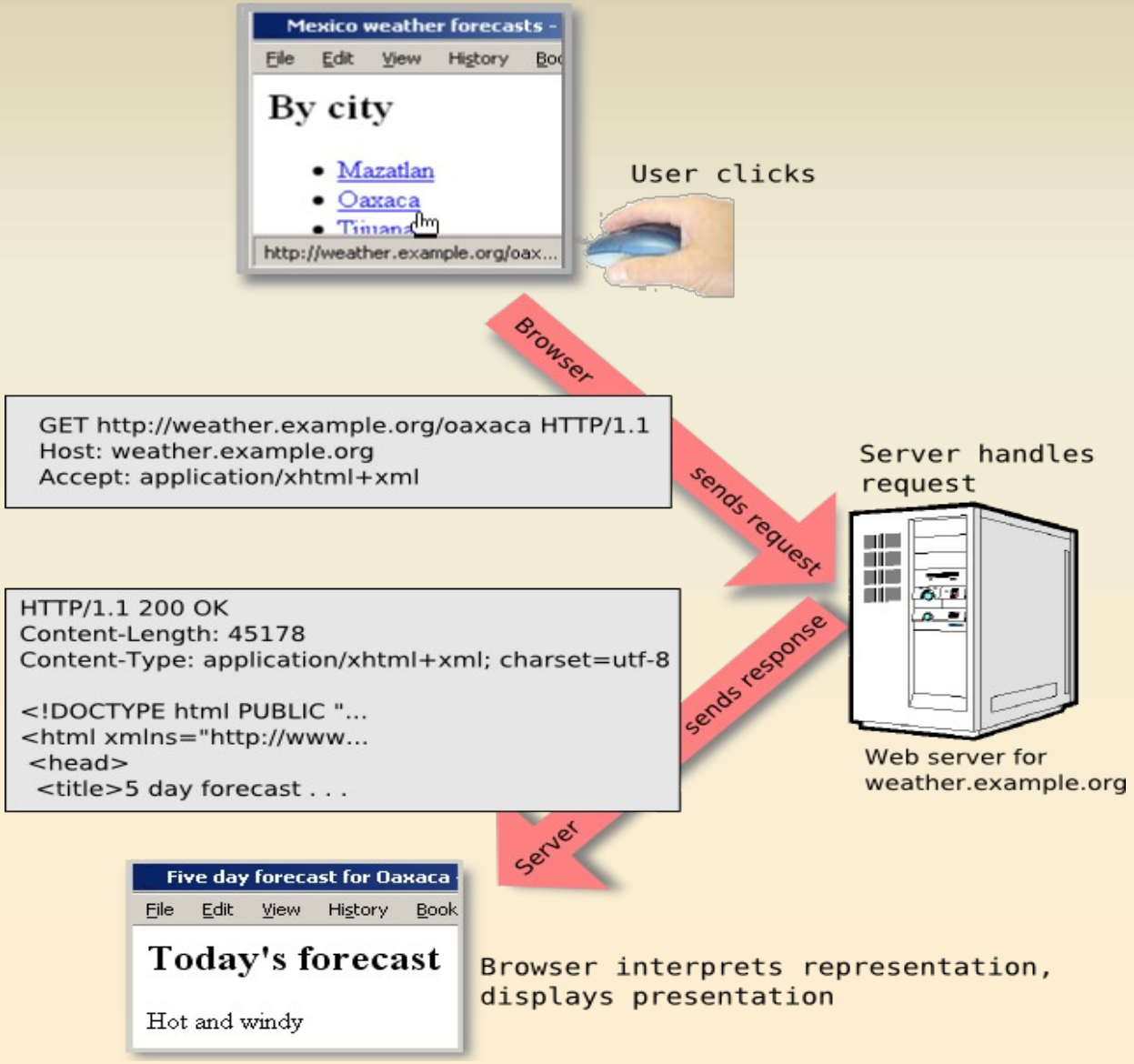
Outline

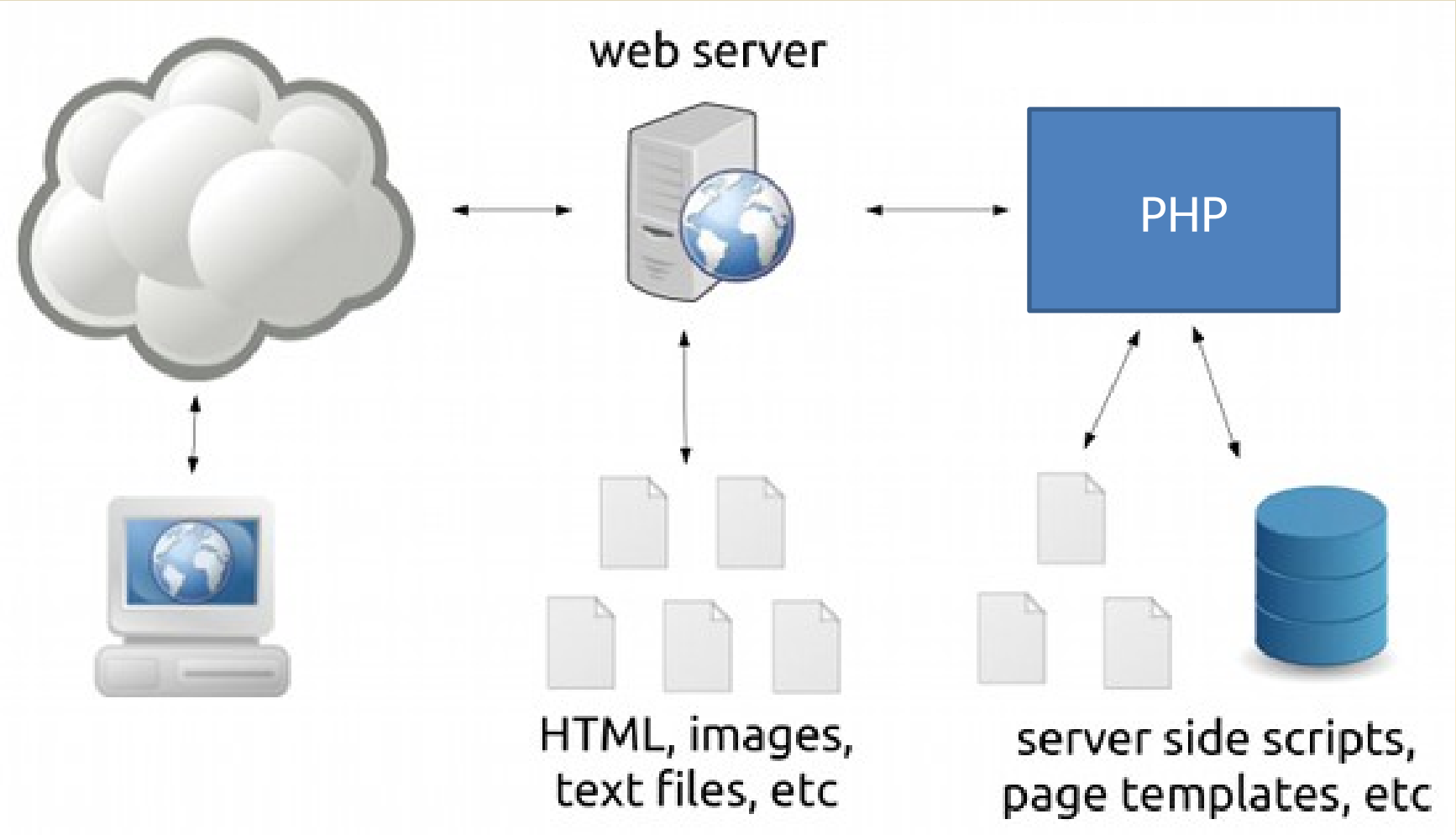
- HTML
- Cascaded Style Sheet (CSS)
- Bootstrap CSS framework
- JavaScript Basic
- PHP and MySQL

So lets start….

How does the Web Works?



How does the Web Works with Dynamic content



Content of a Web Page

- Sound
- Video
- Layers
- Frames
- Forms
- Text
- Lists
- Hyperlinks
- Colour
- Images
- Tables

How does browser display the page?

- All Web pages contain instructions for display
- The browser displays the page by reading these instructions.
- The most common display instructions are called HTML tags.
- HTML tags look like this
  <h1>This is a heading</h1>.

HTML

- **HTML** stands for **H**yper **T**ext **M**arkup **L**anguage
- HTML is not a programming language, it is a **markup language**
- A markup language is a set of **markup tags**
- HTML uses **markup tags** to describe web pages

HTML Markup Tags

- HTML tags are keywords surrounded by **angle brackets** like <html>
- HTML tags normally **come in pairs** like
- and
- The first tag in a pair is the **start tag,** the second tag is the **end tag**
- Start and end tags are also called **opening tags** and **closing tags**

Tags

<!-- ... -->
<!doctype>
<a>
<abbr>
<acronym>
<address>
<applet>
<area>
<b>
<base>
<basefont>
<bdo>
<big>
<blockquote>
<body>

<button>
<caption>

<cite>
<code>
<col>
<colgroup>
<dd>
<del>
<dfn>
<dir>

<dl>
<dt>

<fieldset>
<font>

Tags

<form>
<frame>
<frameset>
<h1>
<h2>
<h3>
<h4>
<h5>
<h6>
<head>
<hr>
<html>
<i>
<iframe>
<img>
<input>
<ins>
<isindex>
<kbd>
<label>
<legend>

<link>
<map>
<menu>
<meta>
<noframes>
<noscript>
<object>

<optgroup>
<option>
<p>
<param>
<pre>
<q>
<s>

Tags

<samp>
<script>
<select>
<small>

<strike>

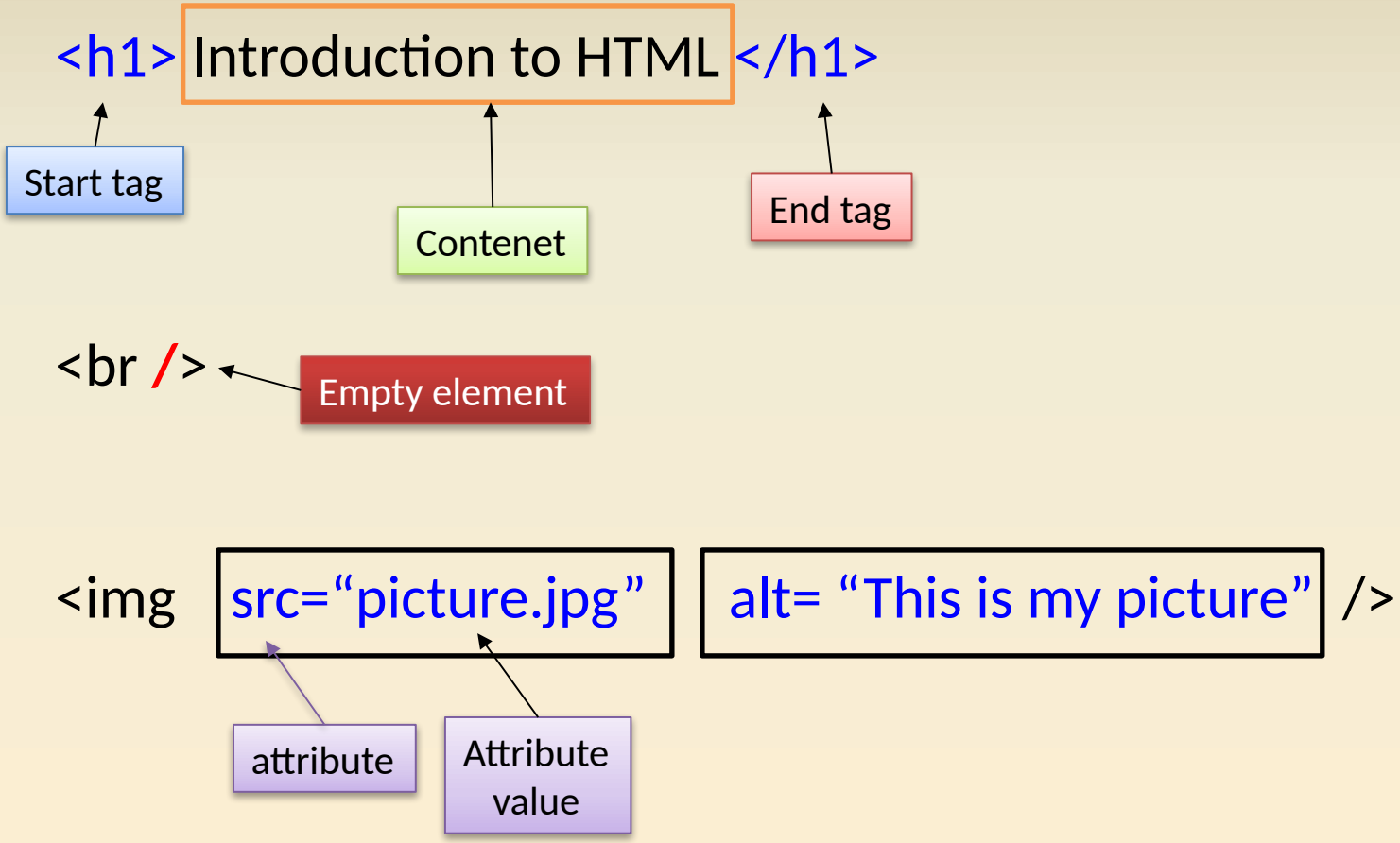
<style>
<sub>
<sup>
<table>
<tbody>
<td>
<textarea>
<tfoot>
<th>
<thead>
<title>
<tr>
<tt>
<u>

<var>

HTML Element Syntax

- Most Elements consist of a start tag, content and an end tag:
  - <h1> Introduction to HTML</h1>
- Empty elements are used to describe elements that do not have any content:
  -
- Attributes are used to describe elements and are placed inside the open tag of an element:
  -
- Comments are used to notate the document, but are not processed by parsers:
  - <!-- This is a comment -->

HTML Element Syntax



Mandatory XHTML Elements

- This is an XHTML document with a minimum of required tags:

```
<!DOCTYPE Doctype goes here>
<html xmlns="http://www.w3.org/1999/xhtml">
      <head>
      <title>Title goes here</title>
      </head>

      <body>
      </body>

</html>
```

!DOCTYPE

- **XHTML 1.0 Strict**

<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Strict//EN"
"http://www.w3.org/TR/xhtml1/DTD/xhtml1-strict.dtd">

XHTML 1.0 Transitional

<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN"
"http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd">

Visit http://24ways.org/2005/transitional-vs-strict-markup/ for more about strict vs transitional doctype.

<body> Tag

- The body element contains all viewable contents of an HTML document, such as
  - text,
  - hyperlinks,
  - images,
  - tables,
  - lists etc

```
<html>
<head>
<title>Title of the document</title>
</head>

<body>
The content of the document……
</body>

</html>
```

<h1> to <h6>

- The <h1> to <h6> tags are used to define HTML headings.
- <h1> defines the largest heading and <h6> defines the smallest heading.
- <h1>This is heading 1</h1>
  <h2>This is heading 2</h2>
  <h3>This is heading 3</h3>
  <h4>This is heading 4</h4>
  <h5>This is heading 5</h5>
  <h6>This is heading 6</h6>

Example

```
<html>
<body>

<h1>This is heading 1</h1>
<h2>This is heading 2</h2>
<h3>This is heading 3</h3>
<h4>This is heading 4</h4>
<h5>This is heading 5</h5>
<h6>This is heading 6</h6>

<p>
Use heading tags only for headi
them just to make something bol
tags for that.
</p>

</body>
</html>
```

This is heading 1

This is heading 2

This is heading 3

This is heading 4

This is heading 5

This is heading 6

Use heading tags only for headings. Don't use them just to make something bold. Use other tags for that.

Tag

- The tag defines a paragraph.
- The p element automatically creates some space before and after itself. The space is automatically applied by the browser, or you can specify it in a style sheet.

This is some text in a paragraph.

```
<html>
<body>

<p>This is a paragraph.</p>
<p>This is a paragraph.</p>
<p>This is a paragraph.</p>

</body>
</html>
```

This is a paragraph.

This is a paragraph.

This is a paragraph.

 Tag

- The
 tag inserts a single line break.
- The
 tag is an empty tag which means that it has no end tag

```
<html>
<body>

<p>
To break<br />lines<br />in a<br
/>paragraph,<br />use the br element.
</p>

</body>
</html>
```

To break
lines
in a
paragraph,
use the br element.

<font> Tag

- The <font> tag specifies the font face, font size, and font color of text.
- Font tag is deprecated in xhtml

```
<font size="3" color="red">This is some text!</font>

<font size="2" color="blue">This is some text!</font>

<font face="verdana" color="green">This is some text!</font>
```

```
<html>
<body>

<p><font size="3" color="red">This is some
text!</font></p>

<p><font size="2" color="blue">This is some
text!</font></p>

<p><font face="verdana" color="green">This is
some text!</font></p>

</body>
</html>
```

This is some text!

This is some text!

This is some text!

Example

<a> Tag

- The <a> tag defines an anchor or hyperlink. An anchor can be used in two ways:
  - To create a link to another document, by using the **href** attribute
  - To create a bookmark inside a document,

```
<a href="http://www.w3schools.com">Visit W3Schools.com!</a>
```

```
<html>
<body>

<a
href="http://www.w3schools.com">Visit
W3Schools.com!</a>

</body>
</html>
```

Visit W3Schools.com!

Various text related tag

- <b> - bold
- - bold
- <i> -italic
- - italic
- <code>- print text in monotype font
- <sub> - subsctipt
- <sup>supersctipt

```
<html>
<body>

<b>This text is bold</b><br />

<strong>This text is strong</strong><br />

<big>This text is big</big><br />

<em>This text is emphasized</em><br />

<i>This text is italic</i><br />

<small>This text is small</small><br />

<tt>This text is teletype</tt><br />

This text contains <sub>subscript</sub><br />

This text contains <sup>superscript</sup>

</body>
</html>
```

This text is bold
This text is strong
This text is big
This text is emphasized
This text is italic
This text is small
`This text is teletype`
This text contains $_{\text{subscript}}$
This text contains ^{superscript}

[Example]

<img> Tag

- The <img> tag embeds an image in an HTML page.
- Images are not technically inserted into an HTML page, images are linked to HTML pages. The <img> tag creates a holding space for the referenced image.

```
<img src="w3schools.jpg" width="104" height="142" alt="logo" />
```

```
<html>
<body>

<img src="angry.gif" alt="Angry face" width="32" height="32"
/>

</body>
</html>
```



Example

<table> Tag

- The <table> tag defines an HTML table.
- A simple HTML table consists of the table element and one or more
  - caption
  - thead
  - tfoot
  - tr,
  - th, and
  - td

ont.)

Some Favorite Albums

Title	Artist
Electric Ladyland	Jimi Hendrix
Revolver	The Beatles
Column 1	Column 2

```
> Electric Ladyland </td>
> Jimi Hendrix </td>
```

```
      <th> Title </th>
      <th> Artist </th>
    </tr>
</thead>
```

```
<tfoot>
   <tr>
      <td> Column 1 </td>
      <td> Column 2 </td>
    </tr>
</tfoot>
```

```
   <tr>
      <td> Revolver </td>
      <td> The Beatles </td>
    </tr>
</tbody>
</table>

</body>
</html>
```

Lists

- ul- unordered list
- ol- ordered list
- dl – definition list

 Tag

- The <ul> tag defines an unordered list (a bulleted list).

-
  Coffee
  Tea
  Milk


```
<html>
<body>

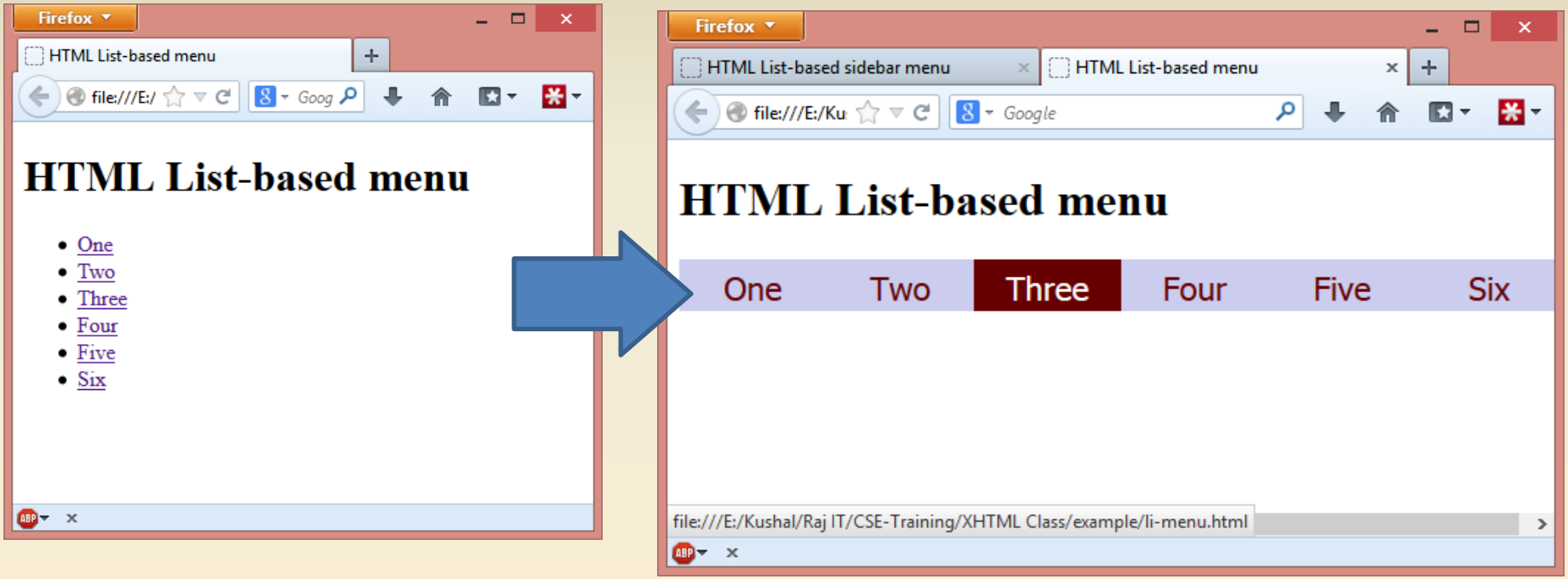
<h4>An Unordered List:</h4>
<ul>
  <li>Coffee</li>
  <li>Tea</li>
  <li>Milk</li>
</ul>

</body>
</html>
```

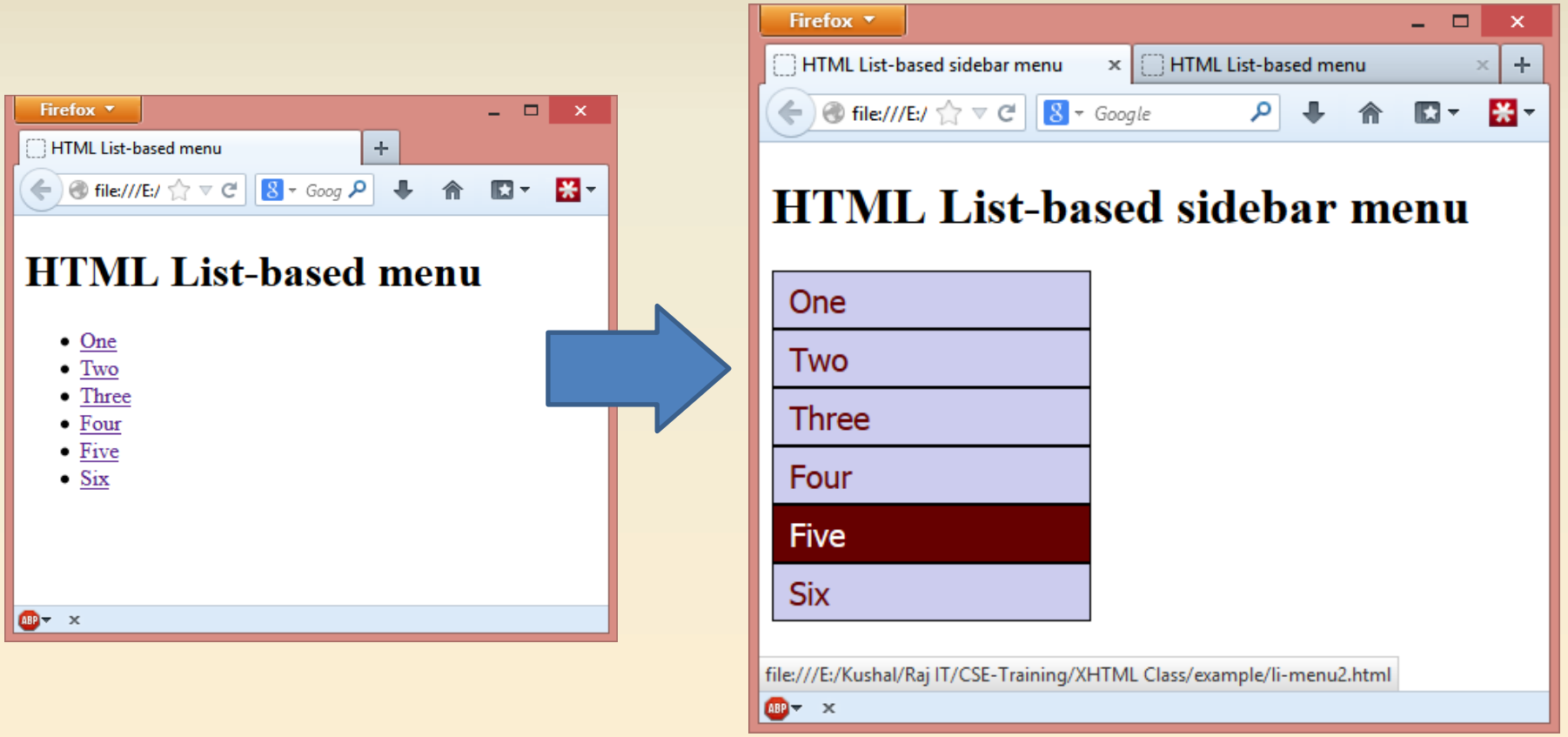
An Unordered List:

- Coffee
- Tea
- Milk

- as top menu



- as sidrbar menu



 Tag

- The <ol> tag is used to create an ordered list.

-
  Coffee
  Tea
  Milk


```
<html>
<body>

<h4>An Ordered List:</h4>
<ol>
  <li>Coffee</li>
  <li>Tea</li>
  <li>Milk</li>
</ol>

</body>
</html>
```

An Ordered List:

1. Coffee
2. Tea
3. Milk

<dl> Tag

- The <dl> tag defines a definition list.
- The <dl> tag is used in conjunction with <dt> (defines the item in the list) and <dd> (describes the item in the list).
- <dl>
  <dt>Coffee</dt>
  <dd>- black hot drink</dd>
  <dt>Milk</dt>
  <dd>- white cold drink</dd>
  </dl>

```
<html>
<body>

<dl>
  <dt>Coffee</dt>
  <dd>- black hot drink</dd>
  <dt>Milk</dt>
  <dd>- white cold drink</dd>
</dl>

</body>
</html>
```

Coffee

- black hot drink

Milk

- white cold drink

Tag

- The tag defines a division or a section in an HTML document.
- The tag is often used to group block-elements to format them with styles.

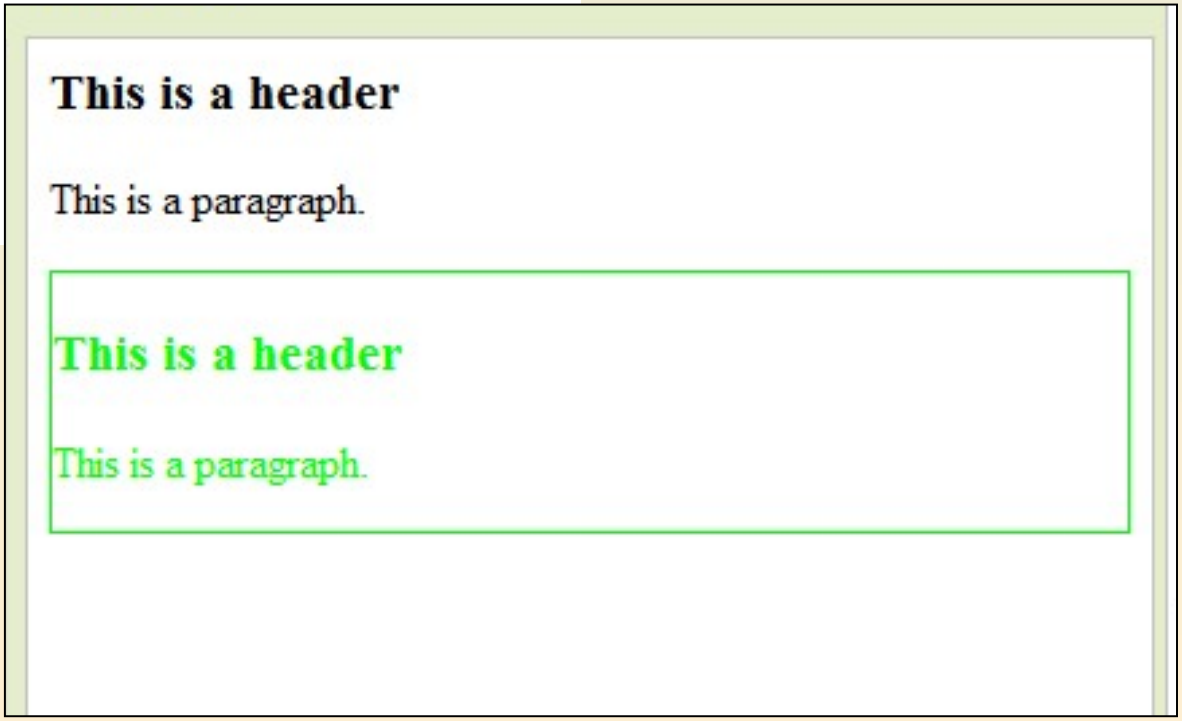
```
<div style="color:#00FF00">
  <h3>This is a header</h3>
  <p>This is a paragraph.</p>
</div>
```

```
<html>
<body>

<h3>This is a header</h3>
<p>This is a paragraph.</p>

<div style="color:#00FF00; border:1px solid">
  <h3>This is a header</h3>
  <p>This is a paragraph.</p>
</div>

</body>
</html>
```



<form> tag

- Used to define an html form

<form **action**=“submit_data.php" **method**="get >

form elements…….

</form>

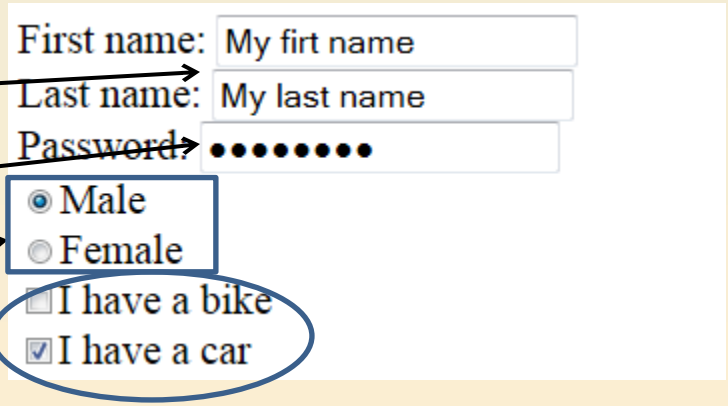
<input /> tag

- Used to define an input field
- Syntex:

<input **type**="text" **name**="my_name" **value**="my_value" />

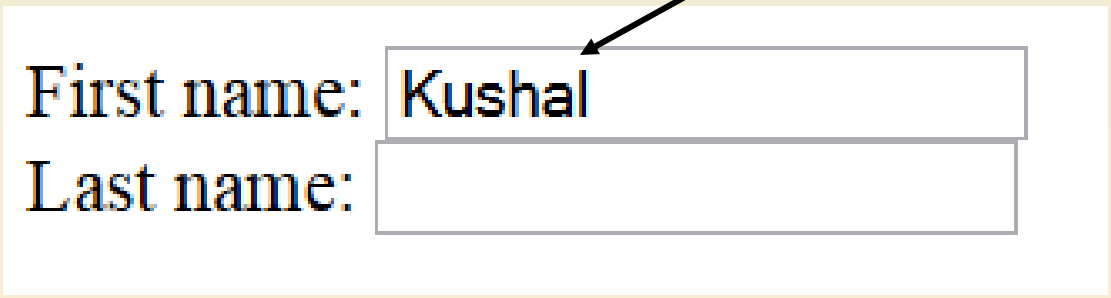
Possible type:

- text
- password
- radio
- checkbox



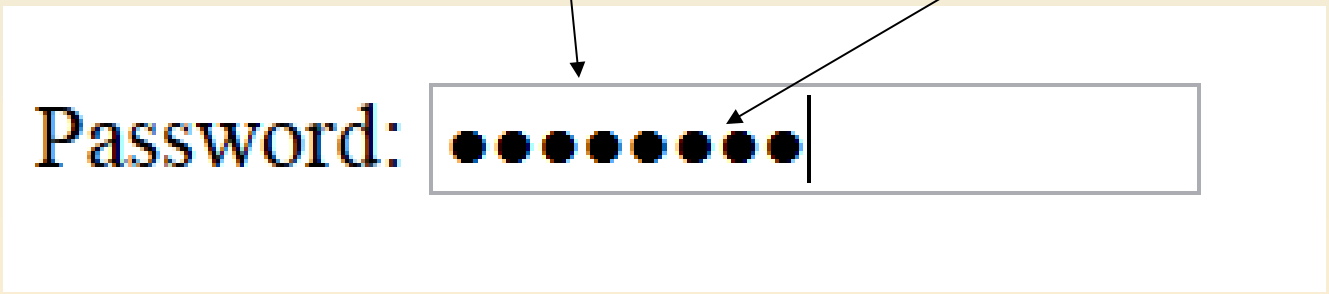
<input>- textbox

```
<form>
First name: <input type="text" name="firstname“ value=“Kushal” />
<br />
Last name:  <input type="text" name="lastname" />
</form>
```



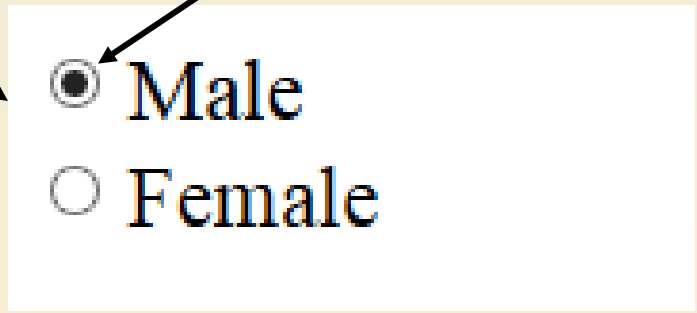
<input>- password

```
<form>
  Password:    <input type="password" name="secret" value="abc123#$" />
</form>
```



<input> - radio

```
<form>
    <input type="radio" name="sex" value="male" checked=“checked” /> Male
  <br />
  <input type="radio" name="sex" value="female" /> Female
</form>
```

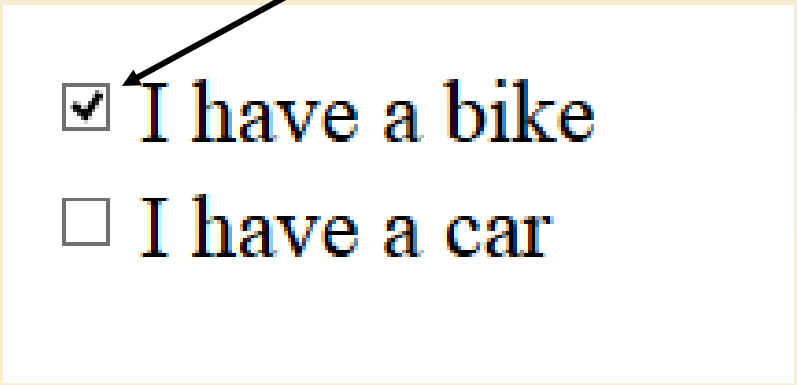


<input> - checkbox

```
<form>

<input type="checkbox" name="vehicle" value="Bike“
checked=“checked” />   I have a bike
<br />
<input type="checkbox" name="vehicle" value="Car" />I have a car

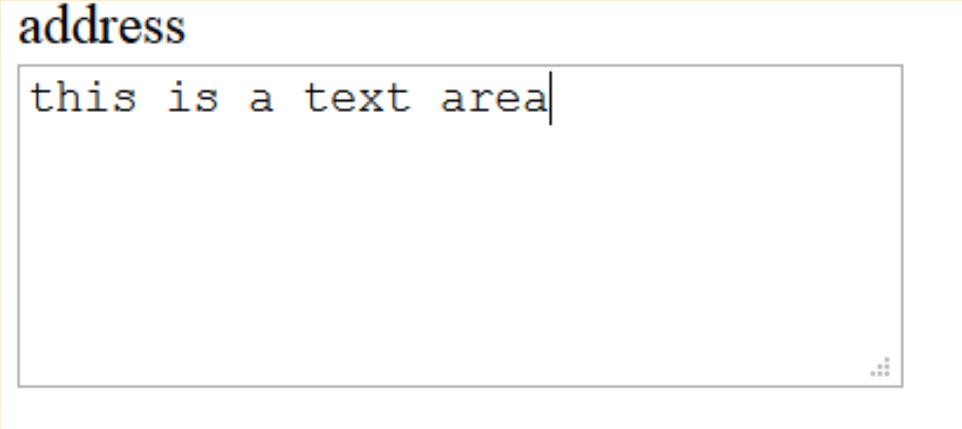
</form>
```



<textarea> tag

- Used to create a multiline textbox

```
<textarea rows="5"  cols="30“  name=“address”>
        this is a text area
</textarea>
```



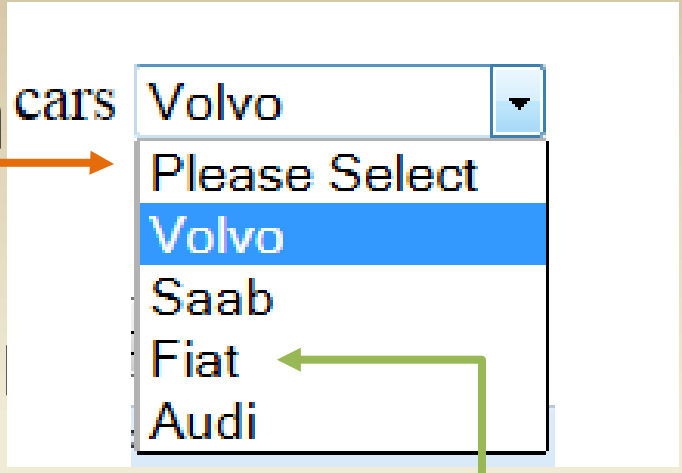
<select> tag

Use 1

- Used to create a dropdown or multi select listbox
- Syntex:

<select name="name of the control" >
  <option value="opt value"> opt label </option>
.
.
.
<option value="opt value n"> opt label n </option>
</select>

<select> ta



Use 1

- Used to create a dropdown combo
- Syntex:

```
<select name="cars" >
    <option value="-1">Please Select</option>
    <option value="volvo">Volvo</option>
    <option value="saab">Saab</option>
    <option value="fiat">Fiat</option>
    <option value="audi">Audi</option>
</select>
```

<select> tag



Use 2

- Used to create a list
- Syntex:

```
<select name="fruits" size="6" >
        <option value="mango">mango</opti
        <option value="Apple">Apple</option>
        <option value="orange">Orange</option>
        <option value="pineapple">Pineapple</option>
</select>
```

<select> tag



Use 3

- Used to create a multi-select list
- Syntex:

```
<select name="fruits" size="6“ multiple="multiple" >
        <option value="mango">mango</option>
        <option value="Apple">Apple</option>
        <option value="orange">Orange</option>
        <option value="pineapple">Pineapple</option>
</select>
```

<submit /> tag

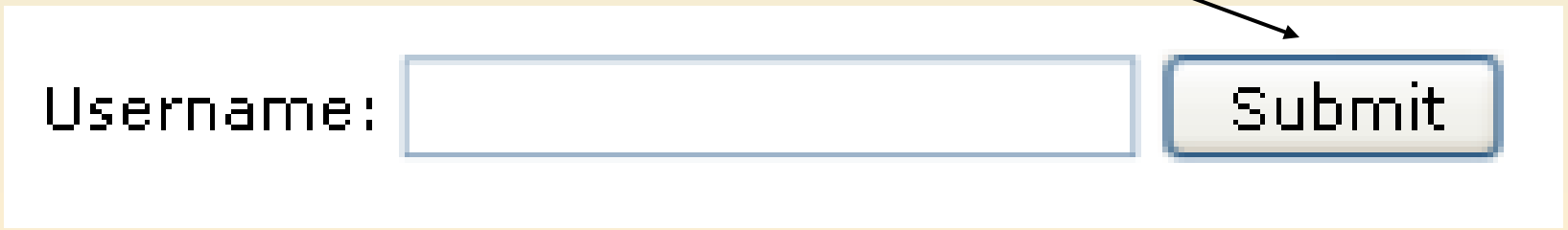
- Used to create a button that will send form data to server when clicked
- Syntax:

```
<input type="submit" value="Send Data" />
```



Forms Element- submit

```
<form action="form_submit.php" method="get">
  Username: <input type="text" name="user" />
  <input type="submit" value="Submit" />
</form>
```



HTML form

- Example

<head> Tag

- The head element is a container for all the head elements. Elements inside <head> can include scripts, instruct the browser where to find style sheets, provide meta information, and more.

- The following tags can be added to the head section:
  - <link>,
  - <meta>,
  - <script>
  - <style>
  - <title>

<style> Tag

- The <style> tag is used to define style information for an HTML document

```
<html>
<head>
  <style type="text/css">

       h1 {  color:red; }

       p { color:blue; font-size:”12px”}

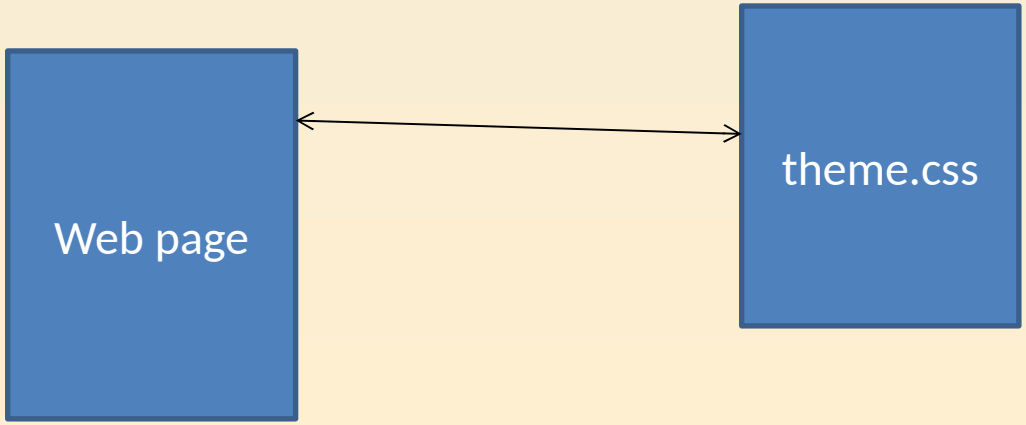
    </style>
  </head>

<body>
       <h1>Header 1</h1>
       <p>A paragraph.</p>
</body>
</html>
```

<link> Tag

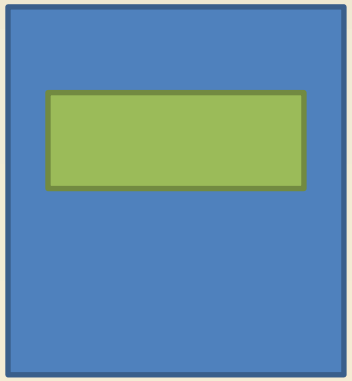
- The <link> tag defines the relationship between a document and an external resource.
- The <link> tag is most used to link to style sheets.

```
<head>
    <link rel="stylesheet" type="text/css" href="theme.css" />
</head>
```



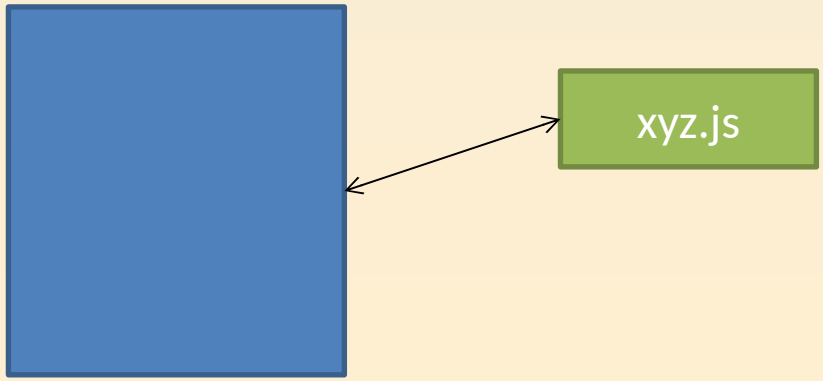
<script> Tag

- The <script> tag is used to define a client-side script, such as a JavaScript.



- ```
  <script type="text/javascript">
  document.write("Hello World!")
  </script>
  ```

- ```
  <script src="xyz.js" type="text/javascript">
  </script>
  ```



<meta> Tag

- The <meta> tag provides meta data about the HTML document.
- Metadata will not be displayed on the page, but it will be used by search engines to index the page in proper category.

- Each meta tag has two attributes
  - name : represents the name of a metadata
  - content : represents the value of the metadata

<meta> Tag

```
<head>
    <meta name="description" content="Free Web tutorials" />
    <meta name="keywords" content="HTML,CSS,XML,JavaScript" />
    <meta name="author" content="Kushal" />
    <meta http-equiv="Content-Type" content="text/html; charset=utf-8" />
</head>
```

- **http-equiv** attribute allows the document author to insert HTTP header information.
- The browser can access this information during read time.

http-equiv

- <meta http-equiv=”**content-type**” content=”charset=***charset***” />
  - Sets the character set for the page
- <meta http-equiv=”**expires**” content=”*date*” />
  - the document will be reloaded after the date is expires.
- **<meta http-equiv="Cache-Control" content="no-cache" />**
  - Used to prevent page caching
- <meta http-equiv=“refresh" content="2” />
  - Used to refresh a page automatically
- <meta http-equiv=“**content-encoding**” content=”*gzip*” />
  - Server sends the page to browser in compressed format

<title> Tag

- The <title> tag defines the title of the document.
- The title element is required in all HTML/XHTML documents.

- ```
  <html>
  <head>
      <title>Title of the document</title>
  </head>
  <body>
  The content of the document......
  </body>
  </html>
  ```
  ```

Differences Between XHTML And HTML

- **XHTML Elements Must Be Properly Nested**

In HTML, some elements can be improperly nested within each other, like this:

```
<b><i>This text is bold and italic</b></i>
```

In XHTML, all elements must be properly nested within each other, like this:

```
<b><i>This text is bold and italic</i></b>
```

Differences Between XHTML And HTML

- **XHTML Elements Must Always Be Closed**

This is wrong:

```
<p>This is a paragraph
<p>This is another paragraph
```

This is correct:

```
<p>This is a paragraph</p>
<p>This is another paragraph</p>
```

Differences Between XHTML And HTML

- **Empty Elements Must Be Closed**

This is wrong:

```
A break: <br>
A horizontal rule: <hr>
An image: <img src="happy.gif" alt="Happy face">
```

This is correct:

```
A break: <br />
A horizontal rule: <hr />
An image: <img src="happy.gif" alt="Happy face" />
```

Differences Between XHTML And HTML

- **XHTML Elements Must Be In Lower Case**

This is wrong:

```
<BODY>
<P>This is a paragraph</P>
</BODY>
```

This is correct:

```
<body>
<p>This is a paragraph</p>
</body>
```

Differences Between XHTML And HTML

- **Some More XHTML Syntax Rules**
  - Attribute names must be in **lower case**
  - Attribute values must be **quoted**
  - The id attribute **replaces** the name attribute
  - The XHTML DTD defines **mandatory** elements

Some advanced topic

HTML Attributes

- HTML elements can have **attributes**
- Attributes provide **additional information** about an element
- Attributes are always specified in **the start tag**
- Attributes come in name/value pairs like: **name="value”**
- Value may be string or numeric
- Some elements don’t have attributes (
)

HTML Standard Attributes

- Core Attributes (**name, class, id, style, title** etc)
- Language Attributes (lang, dir etc)
- Keyboard Attributes (accesskey, tabindex etc)
- Event Attributes
  - Form Events (onblur, onfocus etc)
  - Keyboard Events (onkeydown, onkeypress etc…)
  - Mouse Events (onclick, ondblclick etc…)

Core Attributes

- Common attributes are applied to all the HTML elements
  - **id**
    - a unique identifier for this element in the document
  - **class**
    - classifies this element into one or more subtypes
  - **lang**
    - specifies the language of this element's content along with the language of any attribute content (for example, that inside the title attribute)

Core Attributes

- **dir**
  - specifies the reading direction for text as left to right (ltr) or right to left (rtl)
- **style**
  - specifies an inline style for this element
- **title**
  - provides extra information about the element
- **xml:lang**
  - specifies the language of the content inside this element in XHTML documents

Event Attributes

- attributes that is used to capture an event, and trigger an action with a client-side scripting language such as JavaScript
  - **onblur**
    captures the moment that this element loses focus
  - **onchange**
    captures the moment when the value of this element is changed
  - **onclick**
    captures a mouse click on the element
  - **ondblclick**
    captures a double mouse click on this element
  - **onfocus**
    captures the moment that this element receives focus, either via the cursor or keyboard navigation

Event Attributes

- attributes that is used to capture an event, and trigger an action with a client-side scripting language such as JavaScript
  - **onkeydown**
    captures the event in which a key pressed down (but not released) while focus is on the element
  - **onkeypress**
    captures a key press while focus is on this element
  - **onkeyup**
    captures the moment at which a pressed key is released while focus is on the element
  - **onload**
    catches the point at which a page has completely loaded
  - **onmousedown**
    captures the pressing down of the mouse button while the cursor is positioned over the element
  - **onmousemove**
    captures a mouse movement within this element
  - **onmouseout**
    captures the movement of a cursor off an area covered by this element

Event Attributes

- attributes that is used to capture an event, and trigger an action with a client-side scripting language such as JavaScript

  - **onmouseover**
    captures the movement of a cursor into an area covered by this element
  - **onmouseup**
    captures the release of a previously pressed mouse button while the cursor is over the element
  - **onreset**
    captures the moment that this form's Reset button is activated
  - **onselect**
    captures the event in which the user selects, or highlights, text in the element
  - **onsubmit**
    captures the moment that the form element to which this attribute is applied is submitted
  - **onunload**
    catches the point at which a page has unloaded

THANK YOU